

Question 1: Evaluate integral $I = \int (3x^2 + 2x + 1) \, dx$ from 0 to 2

Step 1: Find antiderivative $F(x) = x^3 + x^2 + x$

Step 2: $F(2) = 2^3 + 2^2 + 2 = 8 + 4 + 2 = 14$

Step 3: $F(0) = 0$

Final Answer: $I = 14 - 0 = 14$

Examiner Note: Checked ✓ [Marks: 5/5]

Question 2: Linear Algebra — Find inverse of matrix $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$

$\det(A) = (2)(3) - (1)(4) = 6 - 4 = 2 \neq 0$

$\text{adj}(A) = \begin{bmatrix} 3 & -1 \\ -4 & 2 \end{bmatrix}$

$A^{-1} = (1/\det(A)) * \text{adj}(A) = 1/2 * \begin{bmatrix} 3 & -1 \\ -4 & 2 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} 1.5 & -0.5 \\ -2 & 1 \end{bmatrix}$

Examiner Note: Checked ✓ [Marks: 5/5]

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Question 3: Solve Differential Equation $dy/dx + 2y = e^{-x}$, $y(0) = 1$

Integrating factor $I(x) = e^{\int 2 \, dx} = e^{2x}$

$d/dx [y * e^{2x}] = e^{2x} * e^{-x} = e^x$

$y * e^{2x} = \int e^x \, dx = e^x + C$

$y(x) = e^{-x} + C * e^{-2x}$

Examiner Mark recorded: 3/5 (Incomplete boundary evaluation?)

Unchecked Intermediate Calculation (Lower Right):
 Boundary condition calculation:
 $y(0) = 1 \Rightarrow 1 = e^0 + C * e^0$
 $1 = 1 + C \Rightarrow C = 0$
 Hence particular solution: $y(x) = e^{-x}$
 [Examiner missed evaluating this working!]

Question 4: Mechanics — Block sliding down frictionless incline of height $h=5\text{m}$

Find final velocity v at the bottom of the incline.

Primary Working:

Using kinematics: $a = g \sin(\theta)$

$$v^2 = u^2 + 2as \Rightarrow v = \sqrt{2 * 9.8 * 5}$$

$$v = \sqrt{98} = 9.9 \text{ m/s}$$

Examiner Mark recorded: 4/5

ROUGH WORK / SCRATCH SPACE (Bottom Margin):

Alternative Energy Conservation Derivation:
 $E_{\text{initial}} = m * g * h, E_{\text{final}} = 0.5 * m * v^2$
 $m * g * h = 0.5 * m * v^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 * 9.8 * 5} = 9.9 \text{ m/s}$
Alternative proof verified energy conservation law explicitly.
[Potentially eligible for bonus reasoning credit]

Question 5: Probability — Bayes' Theorem for Medical Test Accuracy

$P(\text{Disease}) = 0.01, P(+|\text{Disease}) = 0.99, P(+|\text{No Disease}) = 0.05$

$P(+) = (0.99)(0.01) + (0.05)(0.99) = 0.0099 + 0.0495 = 0.0594$

$P(\text{Disease}|+) = 0.0099 / 0.0594 = 0.1667 \text{ (16.67\%)}$

Examiner Note: Checked ✓ [Marks: 5/5]

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[END OF ANSWER BOOKLET]

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